

Integration:

Def. A function $F(x)$ is called antiderivative of a function $f(x)$ on a given open interval if $F'(x) = f(x)$

Example:

- Is $F(x) = \frac{1}{3}x^3$ an antiderivative of $f(x) = x^2$?
Yes, $F'(x) = x^2 = f(x)$.
 - Is $G(x) = \frac{1}{3}x^3 + 5$ an antiderivative of $f(x) = x^2$?
Yes, $G'(x) = x^2 = f(x)$.
 - Is $H(x) = \frac{1}{3}x^6 + 2$ an antiderivative of $f(x) = x^2$?
No, $H'(x) = 2x^5 \neq f(x)$.
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Theorem If $F(x)$ is any antiderivative of $f(x)$ on an open interval, then for any constant $C \in \mathbb{R}$, the function $G(x) = F(x) + C$ is also an antiderivative of $f(x)$.

i.e. $\frac{d}{dx} [F(x) + C] = F'(x) = f(x)$

Integration 

$$\int f(x) dx = F(x) + C.$$

- any function has infinite number of antiderivatives.

$$\frac{d}{dx} \left[\int f(x) dx \right] = f(x).$$

$$\int \frac{d}{dx} [F(x)] dx = F(x) + C.$$

Integration Formulas :

1. $\int dx = x + C.$
2. $\int k dx = kx + C$
3. $\int x^k dx = \frac{x^{k+1}}{k+1} + C$
4. $\int (ax+b)^k dx = \frac{(ax+b)^{k+1}}{(k+1)a} + C.$
5. $\int \frac{1}{x} dx = \ln|x| + C$
6. $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$
7. $\int e^x dx = e^x + C$
8. $\int e^{(ax+b)} dx = \frac{1}{a} e^{(ax+b)} + C$
9. $\int k^{ax+b} dx = \frac{k^{ax+b}}{a \ln k} + C$
10. $\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$
11. $\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$
12. $\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$
13. $\int \csc^2(ax+b) dx = -\frac{1}{a} \cot(ax+b) + C$
14. $\int \sec(ax+b) \tan(ax+b) dx = \frac{1}{a} \sec(ax+b) + C$
15. $\int \csc(ax+b) \cot(ax+b) dx = -\frac{1}{a} \csc(ax+b) + C.$

Example: Evaluate:

$$1// \int 5 dx = 5x + C.$$

$$2// \int x^2 dx = \frac{1}{3} x^3 + C$$

$$3// \int \sqrt{x} dx = \int x^{\frac{1}{2}} dx = \frac{2}{3} x^{\frac{3}{2}} + C$$

$$4// \int \frac{3x^2}{x^3+5} dx = \ln|x^3+5| + C.$$

$$5// \int \frac{x}{x^2-7} dx = \frac{1}{2} \int \frac{2x}{x^2-7} dx = \frac{1}{2} \ln|x^2-7| + C$$

$$6// \int e^3 dx = e^3 x + C$$

$$7// \int e^{3x} dx = \frac{1}{3} e^{3x} + C$$

$$8// \int \sin 7x dx = -\frac{1}{7} \cos 7x + C.$$

$$9// \int \frac{\cos x}{\sin^2 x} dx = \int \csc x \cot x dx = -\csc x + C.$$

$$10// \int (x^7 + \frac{1}{x}) dx = \frac{1}{8} x^8 + \ln|x| + C$$

$$11// \int \left(\frac{1}{2x-1} + (2x+1)^3 - \frac{2x+e^x}{x^2+e^x} \right) dx = \frac{1}{2} \ln|2x-1| + \frac{1}{8} (2x+1)^4 - \ln|x^2+e^x| + C$$

$$12// \int (e^{2x-1} + 5 \sec^2 x - 1) dx = \frac{1}{2} e^{2x-1} + 5 \tan x - x + C.$$

$$13// \int \frac{t^2 - 2t^4}{t^4} dt = \int (t^{-2} - 2) dt = \frac{t^{-1}}{-1} - 2t + C = -\frac{1}{t} - 2t + C.$$

$$14// \int \sec x (\sec x + \tan x) dx = \int (\sec^2 x + \sec x \tan x) dx = \tan x + \sec x + C$$

$$\int k f(x) dx = k \int f(x) dx$$

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

$$\underline{16} \int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx = \tan x - x + C.$$

$$\underline{17} \int \cot^2 x \, dx = \int (\csc^2 x - 1) \, dx = -\cot x + C.$$

$$\underline{18} \int \sin^2 x \, dx = \int \left(\frac{1}{2} - \frac{1}{2} \cos 2x \right) \, dx = \frac{1}{2} x - \frac{1}{4} \sin 2x + C$$

$$\underline{19} \int \cos^2 x \, dx = \int \left(\frac{1}{2} + \frac{1}{2} \cos 2x \right) \, dx = \frac{1}{2} x + \frac{1}{4} \sin 2x + C$$

$$\underline{20} \int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx = -\ln |\cos x| + C = \ln |\sec x| + C.$$

$$\underline{21} \int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx = \ln |\sin x| + C.$$

$$\underline{22} \int \frac{\sin 2x}{\sin x} \, dx = \int \frac{2 \sin x \cos x}{\sin x} \, dx = 2 \sin x + C$$

$$\underline{23} \int x^3 \sqrt{x} \, dx = \int x^{\frac{7}{2}} \, dx = \frac{2}{9} x^{\frac{9}{2}} + C.$$

$$\underline{24} \int (2+y^2)^2 \, dy = \int (4 + 4y^2 + y^4) \, dy = 4y + \frac{4}{3} y^3 + \frac{1}{5} y^5 + C$$

Note that:

$$\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x$$

$$\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x$$

$$\tan^2 x = \sec^2 x - 1$$

$$\cot^2 x = \csc^2 x - 1$$

$$\sin 2x = 2 \sin x \cos x.$$

Example

1. Find $f(x)$ if $f'(x) = \sin x + 1$ where $f(\frac{\pi}{3}) = \frac{1}{2}$.

sol.

$$f(x) = \int (\sin x + 1) dx = -\cos x - x + C$$

Now,

$$f(\frac{\pi}{3}) = -\frac{1}{2} - \frac{\pi}{3} + C = \frac{1}{2} \longrightarrow C = 1 - \frac{\pi}{3}.$$

$$\therefore f(x) = -\cos x - x + 1 - \frac{\pi}{3}.$$

2. Find $f(x)$ if $f'(x) = 4e^x$ where $f(0) = 1$.

sol.

$$f(x) = \int 4e^x dx = 4e^x + C.$$

$$f(0) = 4 + C = 1 \longrightarrow C = -3$$

$$\therefore f(x) = 4e^x - 3.$$

3. Find $f(x)$ if $f''(x) = x + \cos x$ where $f(0) = 2$ and $f'(0) = 1$.

sol.

$$f'(x) = \int (x + \cos x) dx = \frac{1}{2}x^2 + \sin x + C_1$$

$$f'(0) = 0 + 0 + C_1 = 1 \longrightarrow C_1 = 1$$

$$\therefore f'(x) = \frac{1}{2}x^2 + \sin x + 1.$$

$$f(x) = \int (\frac{1}{2}x^2 + \sin x + 1) dx = \frac{1}{6}x^3 - \cos x + x + C_2$$

$$f(0) = 0 - 1 + 0 + C_2 = 2 \longrightarrow C_2 = 3$$

$$\therefore f(x) = \frac{1}{6}x^3 - \cos x + x + 3.$$

• Integration by Substitution :

$$\int f(g(x)) g'(x) dx$$

$$\rightarrow \text{Let } y = g(x)$$

$$dy = g'(x) dx$$

$$\frac{dy}{g'(x)} = dx$$

$$\int f(g(x)) g'(x) dx = \int f(y) g'(x) \cdot \frac{dy}{g'(x)}.$$

$$= \int f(y) dy$$

$$= F(y) + C = F(g(x)) + C.$$

Example: Evaluate:

$$\begin{aligned} 1 // \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx &\rightarrow \text{let } y = \sqrt{x} \\ dy &= \frac{1}{2\sqrt{x}} dx \\ 2\sqrt{x} dy &= dx \end{aligned}$$

$$\begin{aligned} &= \int \frac{\sin y}{\sqrt{x}} \cdot 2\sqrt{x} dy = -2 \cos y + C. \\ &= -2 \cos \sqrt{x} + C. \end{aligned}$$

$$\begin{aligned} 2 // \int \frac{dx}{x(\ln x)^2} &\rightarrow \text{let } y = \ln x \\ dy &= \frac{1}{x} dx \\ x dy &= dx \end{aligned}$$

$$= \int \frac{x dy}{x y^2} = \frac{-1}{y} + C = \frac{-1}{\ln x} + C$$